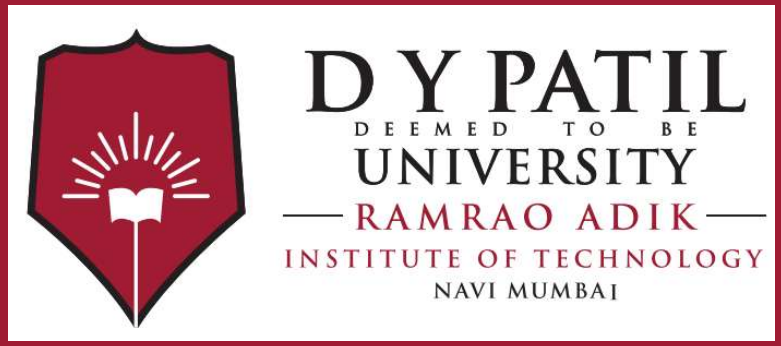


Subject Name: OOP

Unit No:1

Unit Name: Introduction to OOP

Index

Lecture 1 – Basic Concepts of Object-Oriented Programming	7
Lecture 2 – Need of OOP, Object Oriented Programming Paradigm	21
Lecture 3 – Benefits of OOP, Introduction to OOP languages : C++ and Java	34



D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Course Schema

Teaching Scheme &Evaluation Scheme:

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	PR / ORAL	Total
231CEUCC32	Object Oriented Programming	03	--	--	03	--	--	03

Course Code	Course Name	Examination Scheme								
		Theory						TW	PR	Total
		Internal Assessment				End Sem Exam	Exam. Duration (in Hrs)			
		Test1	Test2	Avg.	Mid Sem Exam					
231CEUCC32	Object Oriented Programming	20	20	20	20	60	02	--	--	100

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Course Schema

Teaching Scheme &Evaluation Scheme:

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	PR / ORAL	Total
231CEUCL32	Object Oriented Programming Lab	--	02	--	--	--	01	01

Course Code	Course Name	Examination Scheme								
		Theory						TW	PR	Total
		Internal Assessment				End Sem Exam	Exam. Duration (in Hrs)			
		Test1	Test2	Avg.	Mid Sem Exam					
231CEUCL32	Object Oriented Programming Lab	--	--	--	--	--	--	25	25	50

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Course Objectives

1. To understand the object-oriented programming basics and its features.
2. Able to use a programming language to resolve problems.
3. To understand and apply Object Oriented Programming (OOP) principles using Java.
4. To study various java programming concept like multithreading, exception handling, packages etc.
5. To explain components of GUI based programming.

D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Course Outcomes

Students will be able to :

1. Understand basics of OOP and apply fundamental programming constructs.
2. Understand and illustrate the features of classes and objects.
3. Elaborate the concept of strings, arrays and vectors.
4. Develop a program that implements the concept of inheritance, interfaces and packages.
5. Implement the notion of exception handling and multithreading.
6. Develop GUI based application.

D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Lecture No: 1
Basic Concepts of Object-Oriented Programming



Introduction to Programming

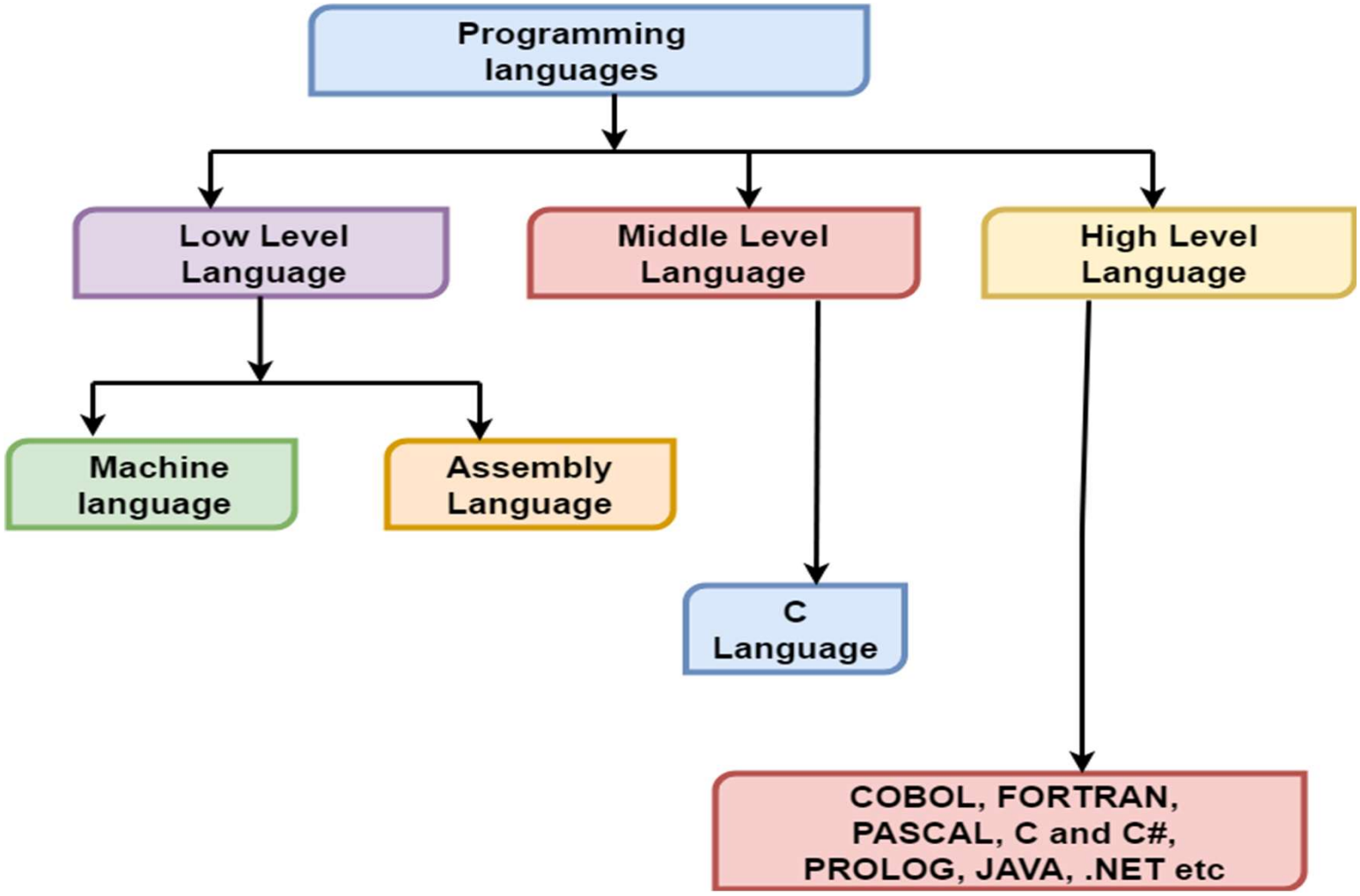
- **Program:**
  - Program (also software or application) is a sequence of instructions written to perform a specified task with a computer.

- **Programmer:**
  - A person who practices this skill of writing program is referred to as a computer programmer or software developer or coder.

- **Programming:**
  - Programming is the iterative process of writing or editing source code. Editing source code involves testing, analyzing, refining and sometimes coordinating with other programmers on a jointly developed program.

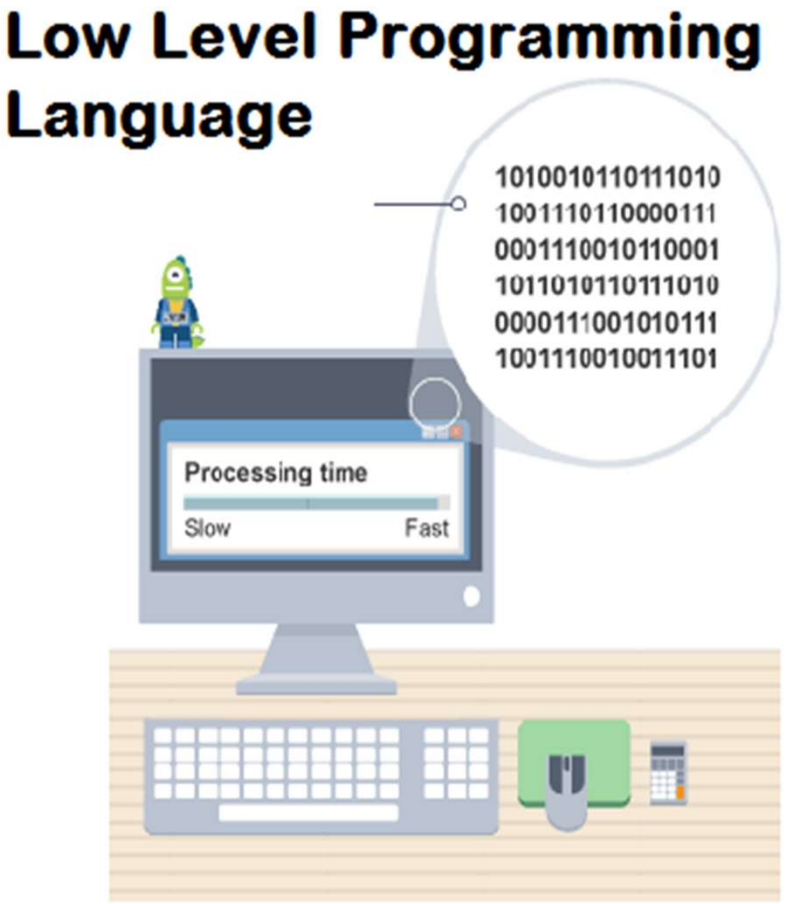
D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

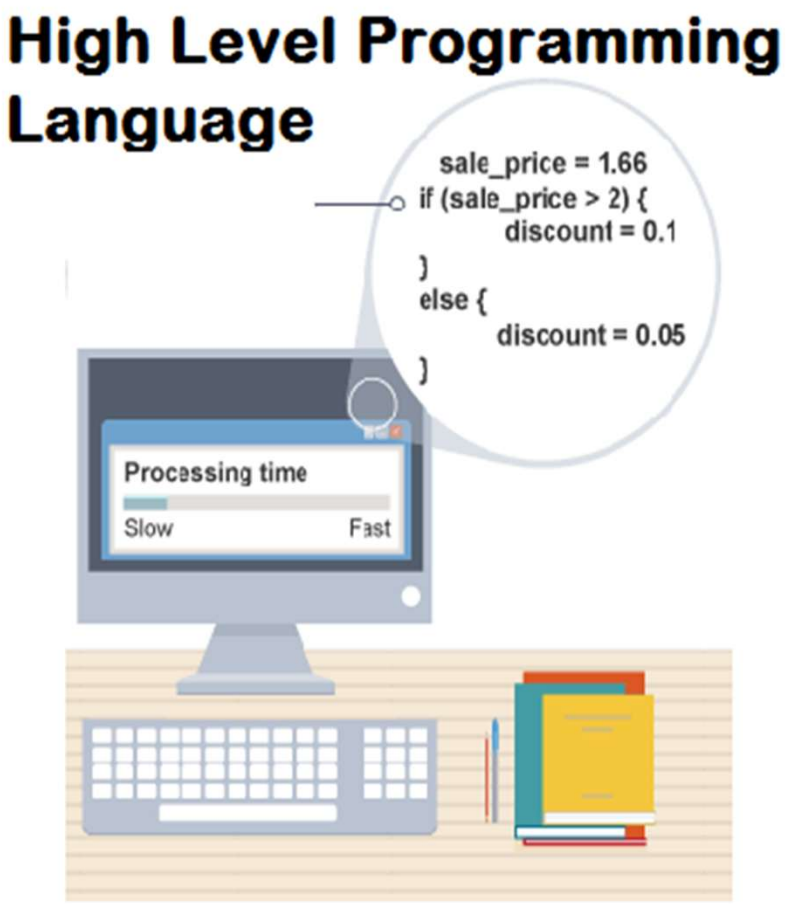
Hierarchy of Programming Languages



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Why to Learn High Level Languages





D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Programming Approaches

- Procedure-Oriented: Top-Down Approach
- Object-Oriented: Bottom-Up Approach

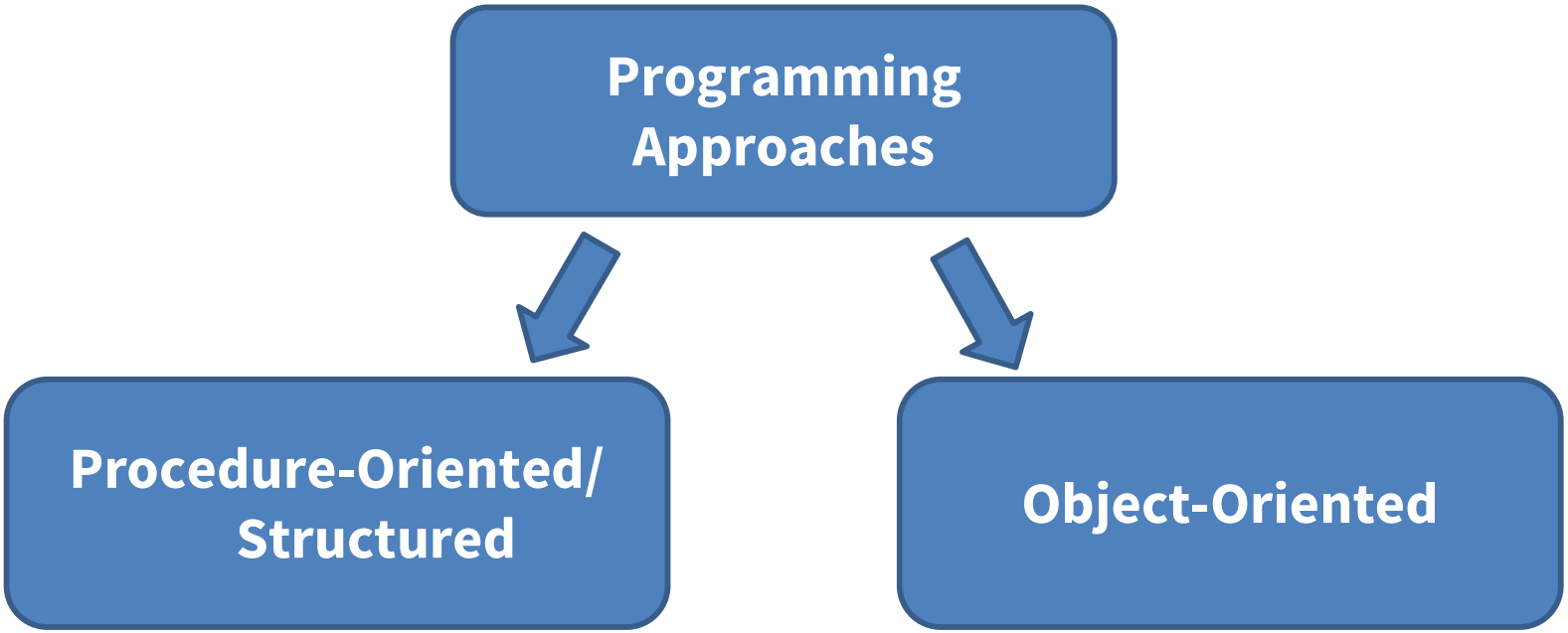


Figure: Programming Approaches

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Procedural Programming

- Traditional programming languages were procedural.
  - E.g. C, Pascal, BASIC, Ada and COBOL
- Programming in procedural languages involves choosing data structures (appropriate ways to store data), designing algorithms, and translating algorithm into code.
- In procedural programming, data and operations on the data are separated.
- This methodology requires sending data to procedure/functions.

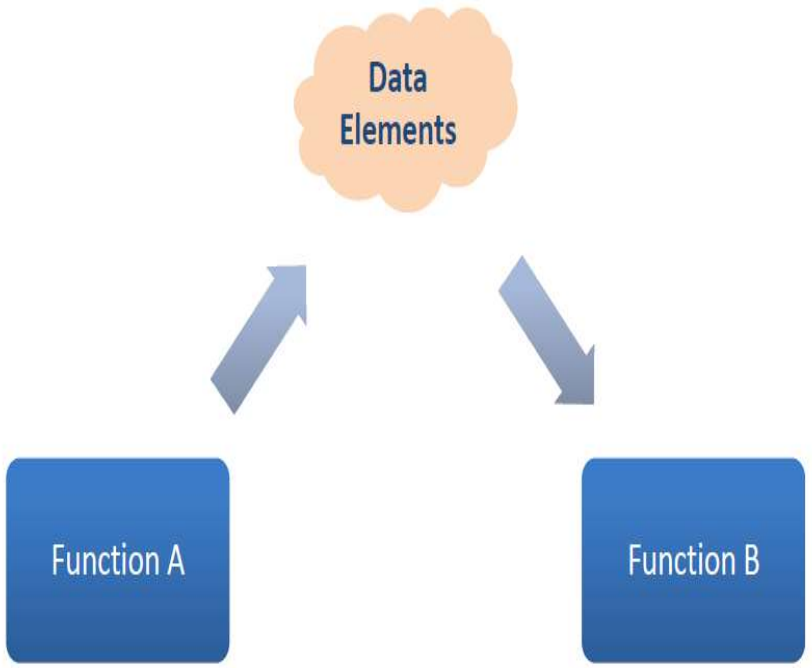
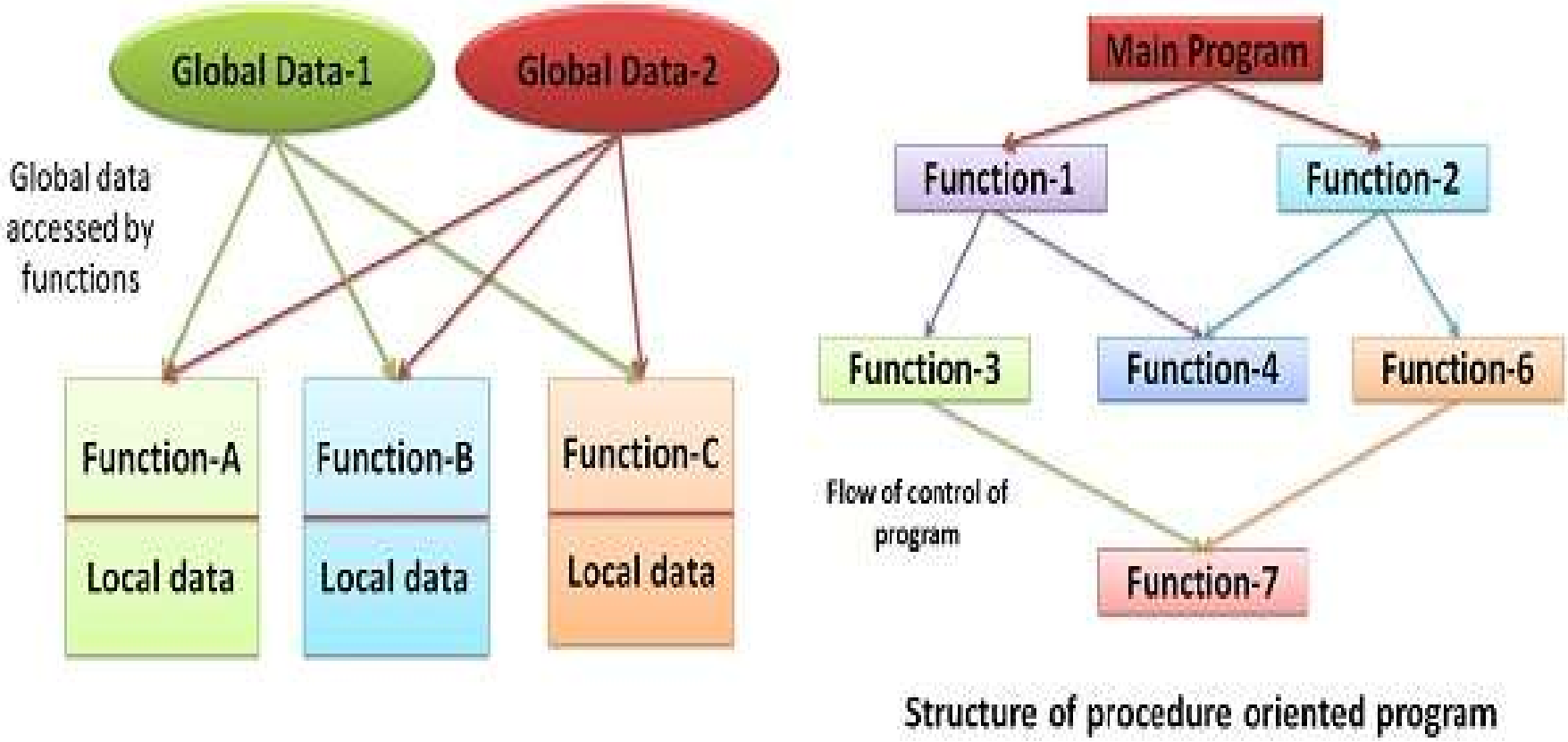


Figure: Procedural Programming

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Structure of Procedure-oriented Programming



Structure of procedure oriented program

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Structured Programming Approach

- The program is made as a single structure i.e. the code will execute the instruction by instruction one after the other.
- It doesn't support the possibility of jumping from one instruction to some other with the help of any statement like GOTO, etc. Therefore, the instructions in this approach will be executed in a serial and structured manner.
- C is called a structured programming language because to solve a large problem, C programming language divides the problem into smaller structural blocks each of which handles a particular responsibility.
- These structural blocks are –
  - Decision making blocks like if-else-elseif, switch-cases,
  - Repetitive blocks like For-loop, While-loop, Do-while loop etc
  - Subroutines/Procedures - functions

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Advantages of Structured Programming

- C structured programming is simple and easy to understand and implement.
- It is well sited for small size implementation. However this is not restricted. A good design can extend it to large size implementation.
- Programmers do not require to know complex design concepts to start a new program.
- Easier to Debug
- These are mainly problem oriented rather than machine based.
- Program written in a higher level language can be translated into many machine languages and therefore can run on any computer for which there exists an appropriate translator.
- It is independent of machine on which it is used i.e. programs developed in high level languages can be run on any computer.

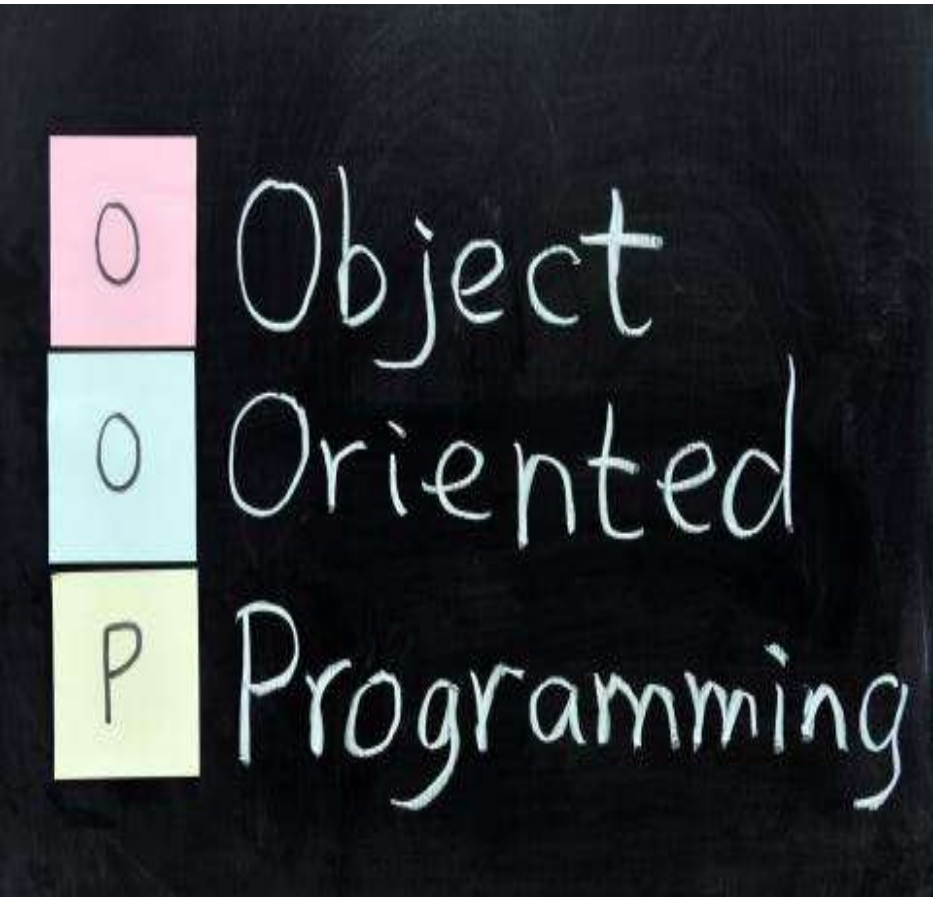
D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Disadvantages of Structured Programming

- A high level language has to be translated into the machine language by translator and thus a price in computer time is paid.
- Since it is Machine-Independent, So it takes time to convert into machine code.
- The code generated by a translator might be inefficient compared to an equivalent assembly language program.
- Data type are proceeds in many functions in a structured program. When changes occur in those data types, the corresponding change must be made to every location that acts on those data types within the program. This is really a very time consuming task if the program is very large.
- Data and methods are not bind together in a module.

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Object Oriented Approach



- Object oriented approach organizes a program in the form of objects
- Focus on data
- Program is organized around data
- Data and operations are grouped together

Objects = Data+ Methods

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Object-Oriented Programming

- Object-oriented programming is centered on creating **objects rather than procedures/ functions.**
- Objects are a melding of **data and procedures that manipulate that data.**

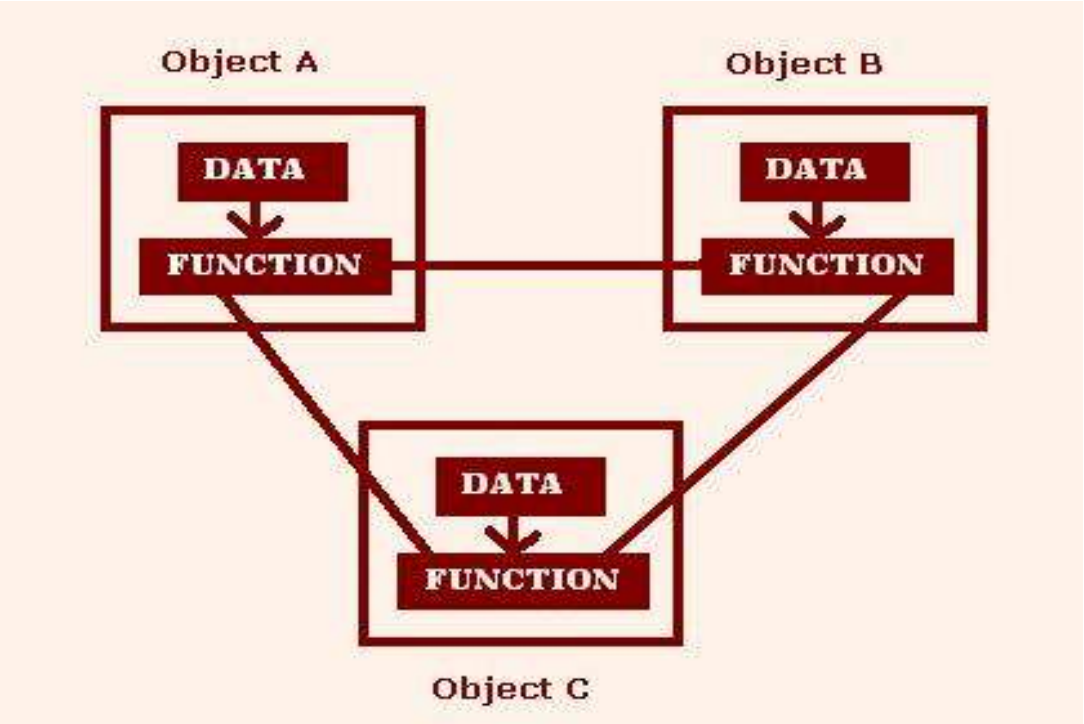
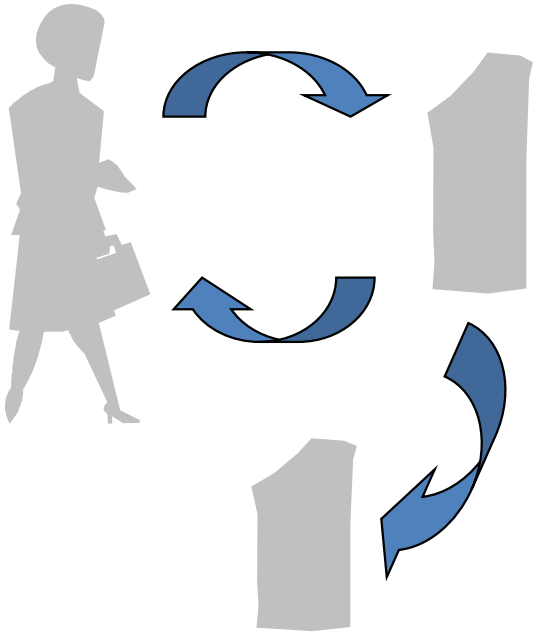


Figure: Object-Oriented Programming

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

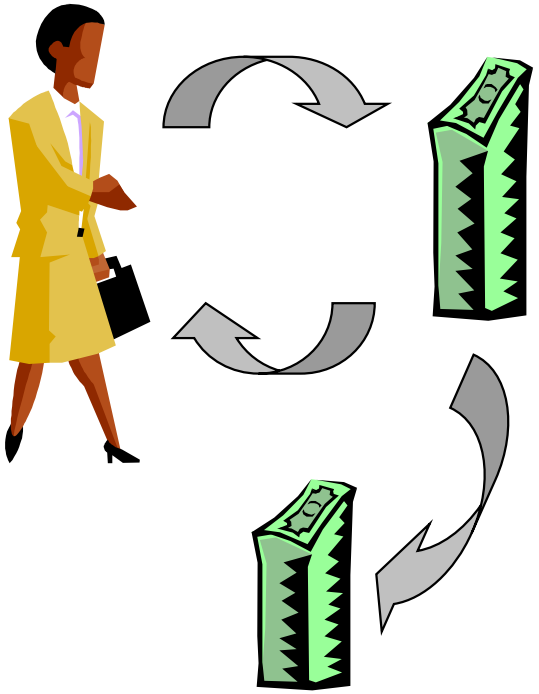
Procedural vs Object oriented programming

- Procedural



Withdraw, deposit, transfer

- Object Oriented



Customer, money, account

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Difference between Structured and Object Oriented Approach

Structured Programming	Object Oriented Programming
• The program is represented as a logical structure.	• The program is written as a collection of objects which communicate with each other.
• The flow of execution of the programming is dependent on the structure of the program.	• The basic entity is object. Each computation is performed using objects only.
• Code is given more importance.	• Data is given more importance.
• Can handle up to moderately complex programs.	• Can handle very complex programs.
• Less data security.	• More data security.
• Less code reusability.	• More reusability.
• Flexibility is less.	• Flexibility is more.
• Abstraction is less.	• Abstraction is more.
• This is top-down approach.	• This is a bottom-up approach.
E.g: C, VB, FORTRAN, Pascal	E.g: C++, JAVA, VB.NET, C#.NET

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Module No 1: Introduction to OOP

Lecture No: 2
Need of OOP, Object Oriented Programming Paradigm



Need of OOP

Hardware	• Language close to hardware
Object Oriented Programming	• Easy to understanding
Software Developers Choice	• Millions of software developers use C++,Java
Application And System Programming.	• C++, Java is the most widely used programming languages
Difference Between Compiler, Linker And Loader	• C++, Java really teaches you the difference between compiler, linker and loader, different data types, storage classes,

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Applications of OOP

Application Software Development	• Operating System: Windows, Mac OSX and Linux. • Browsers : Mozilla Firefox and Chrome • database system: MySQL.
Programming Languages Development	• C#, Java, JavaScript, Perl, UNIX's C Shell, PHP and Python, and Verilog etc.
Computation Programming	• Has a fast speed and computational efficiencies.
Embedded System	• Medical and Engineering Applications like softwares for MRI machines, high-end CAD/CAM systems etc.
Games Development	• Procedural programming for CPU intensive functions and provides greater control over hardware

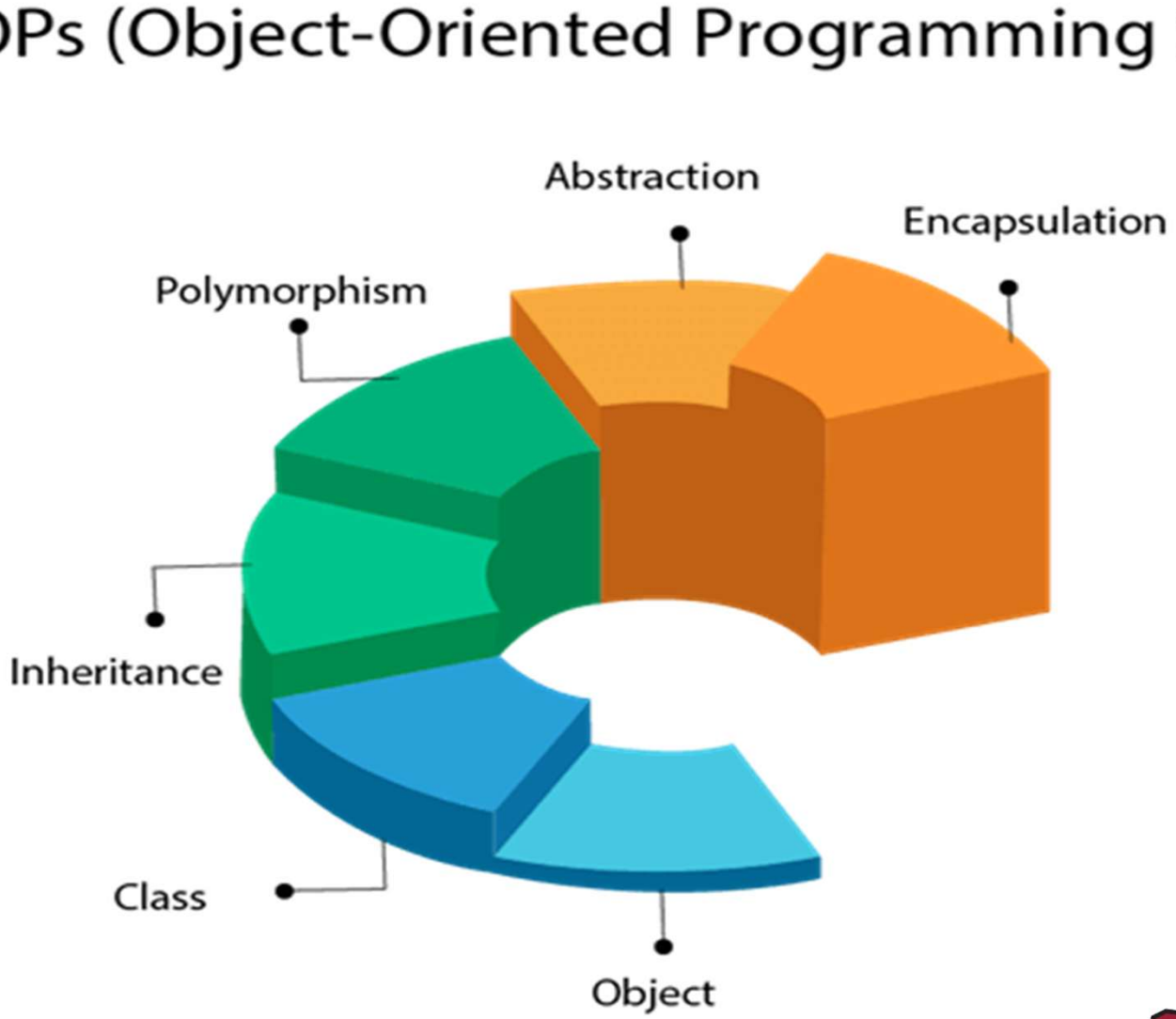
D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Features of OOP

Striking features of OOP	
	Emphasis data
	Program divided into object
	Data structure
	Function
	Hidden Data
	Object Communication
	New data/function easily add
	Follow bottom of approach

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Object Oriented Paradigm



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Object Oriented Programming

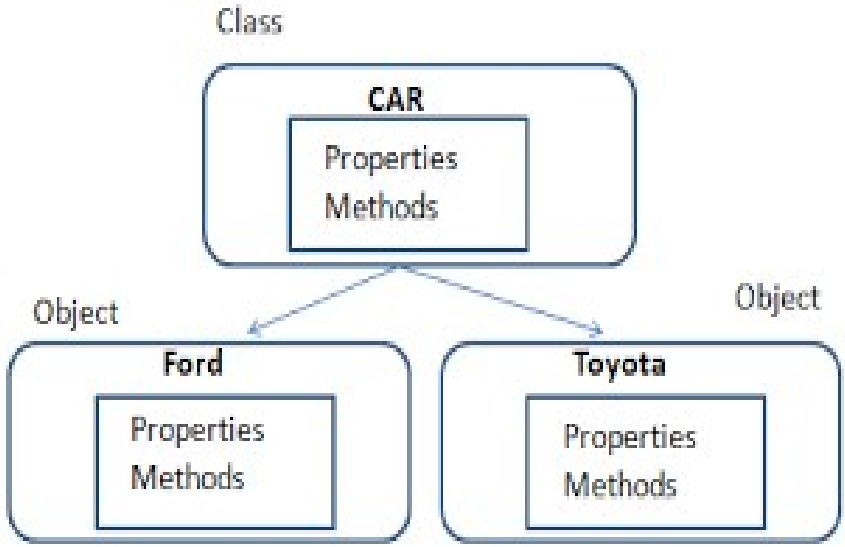
- Classes
- Objects
- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Dynamic Binding
- Message Passing

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Classes and Objects

➢ Classes

- Basic building blocks OOP and a single entity which has data and operations on data together.
- Classes are user-defined data types and behave like the built-in types of a programming language.



➢ Objects

- Objects are the basic run-time entities.
- The instances of a class which are used in real functionality(properties and methods)
- When a program is executed the objects interact by sending messages to one another.

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Abstraction

- Abstraction refers to the act of representing essential features without including the background details or explanations.
- Specifying what to do but not how to do.
- It is a flexible feature for having a overall view of an object's functionality.
- Classes use the concept of abstraction and are defined as a list of abstract attributes such as size, weight and cost, and functions to operate on these functions.
- Since the classes use the concept of data abstraction, they are known as Abstract Data Types.

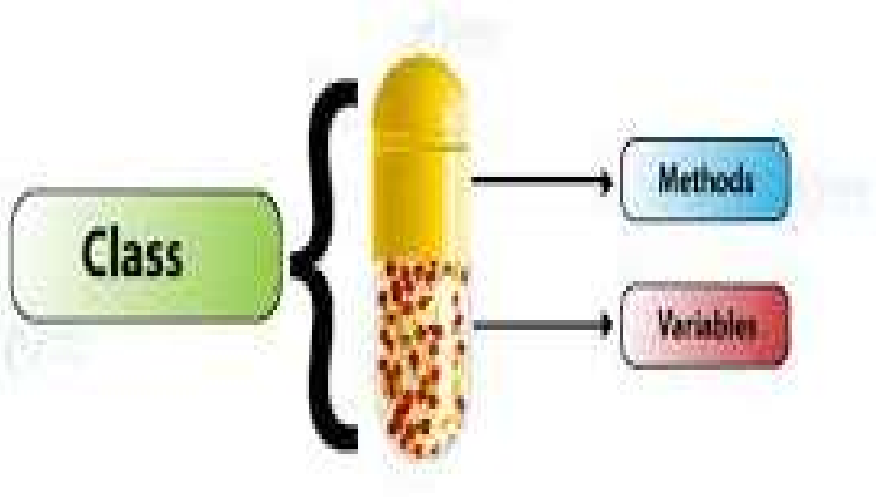


Real Life Example of Abstraction

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Encapsulation

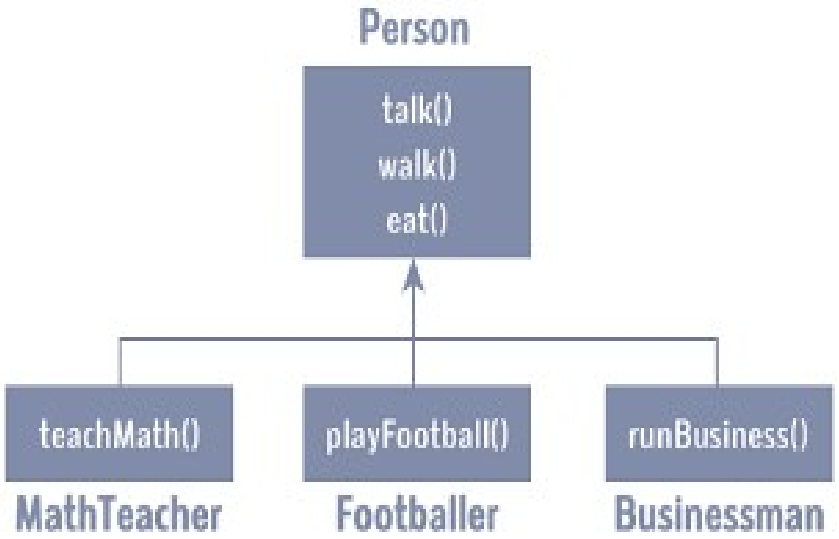
- The wrapping up of data and functions into a single unit is known as Encapsulation.
- The data is not accessible to the outside world and only those functions which are wrapped in the class can access it.
- This insulation of the data from direct access by the program is called data hiding or information hiding.



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Inheritance

- Inheritance is the process by which objects of one class acquire the properties of objects of another class.
- The principle behind this sort of division is that each derived class shares common characteristics with the class from which it is derived.
- In OOP the concept of inheritance provides the idea of reusability.
- This means that we can add additional features to an existing class without modifying it.

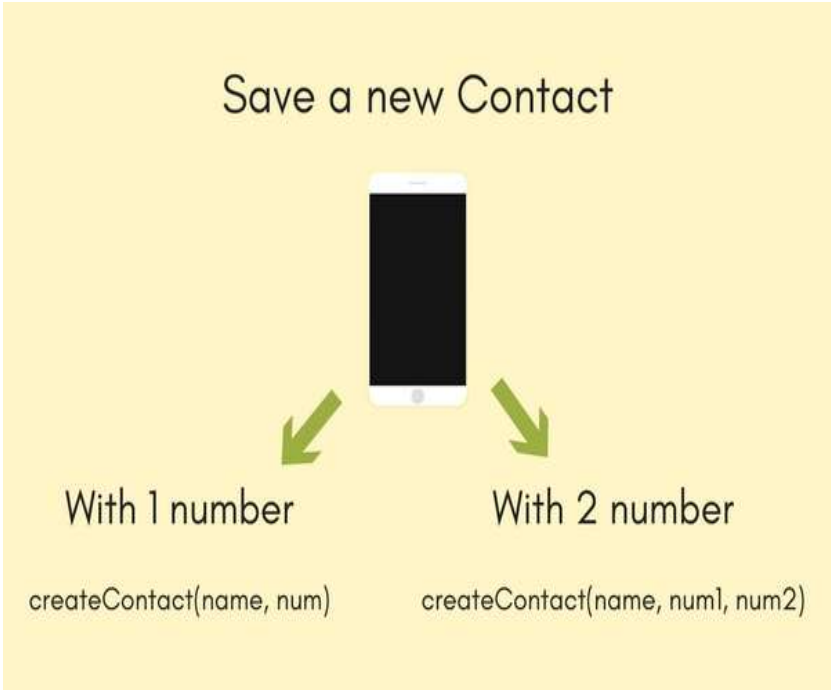




D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Polymorphism

- Polymorphism means the ability to take more than one form.
- Multiple definitions for a single name.
- Functions with same name with different functionality
- That is an operation may exhibit different behaviours in different instances.
- saves time in investing many function names
- The behaviour depends upon the types of data used in the operation.
- Operator and Function overloading
- Example: Operation of addition.



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

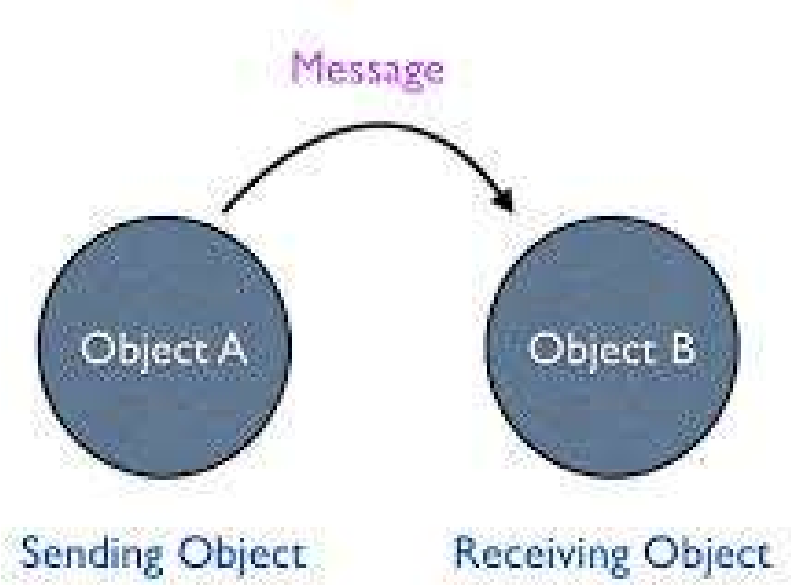
Dynamic Binding

- Dynamic binding refers to the linking of a procedure call to a specific sequence of code.
- The code to be executed for a specific procedure call is not known until run-time.
- Dynamic dispatch is generally used when multiple classes contain different implementations of the same method.
- It provides a mechanism for selecting the function to be executed from various function alternatives at the run-time.
- It is associated with polymorphism and inheritance.

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Message Communication

- Object communicate with one another by sending and receiving information much the same way as people pass message to one another.
- Objects communicates through invoking methods and sending data to them.
- This feature of sending and receiving information among objects through function parameters is known as Message Passing.



Message Passing

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Lecture No: 3
Benefits of OOP, Introduction to OOP languages : C++ and Java



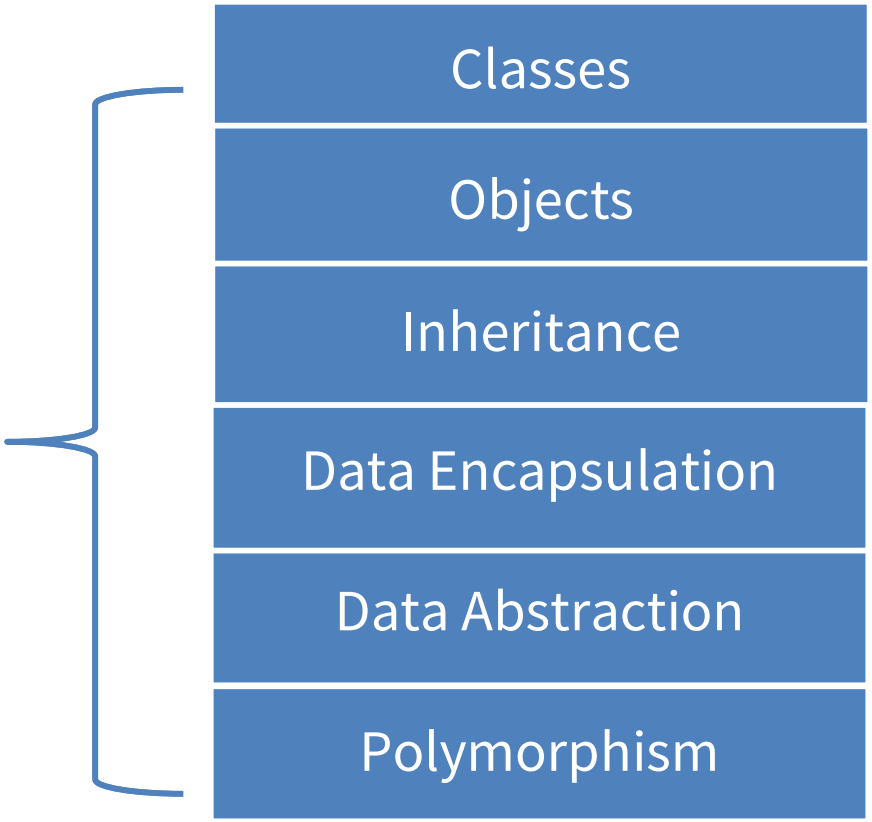
Benefits of OOP



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

OOP: C++ Language

- Developed by Bjarne Stroustrup in 1979.
- an extension / superset of C language.
- Supports both procedural and object oriented paradigm
- The prime purpose of C++ programming was to add **object orientation** to the C programming language.
- While designing C++ modules, we try to see whole world in the form of objects.



D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Difference Between C and C++

C	C++
C is the function or procedure programming language.	C++ is the object oriented programming language.
Top down approach is used in C.	Bottom up approach is used in C++.
C is less secure than C++	C++ is secure, because oop's concept is available. Here Data is hidden and can't be accessed by external functions.
In C programming variable declaration is possible only top of program.	In C++ you can any where declare variable.
In C, namespace feature is not available.	In C++, namespace feature is present.
C is a middle level language.	C++ is a high level language.
In this programming, programs are divided into modules and functions.	In this programming, programs are divided into classes and functions.
Here no exception handling concept is available	It support exception handling concept.

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

OOP: Java Language

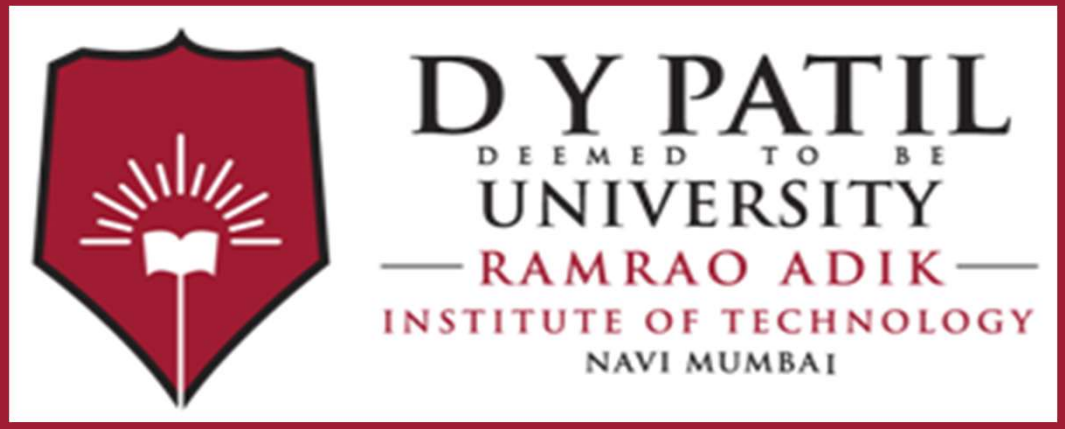
- When we consider a Java program, it can be defined as a collection of objects that communicate via invoking each other's.
- **Object** − Objects have states and behaviors. Example: A dog has states - color, name, breed as well as behaviors - wagging, barking, eating. An object is an instance of a class.
- **Class** − A class can be defined as a template/blueprint that describes the behaviors/states that object of its type support.
- **Methods** − A method is basically a behavior. A class can contain many methods. It is in methods where the logics are written, data is manipulated and all the actions are executed.
- **Instance Variables** − Each object has its unique set of instance variables. An object's state is created by the values assigned to these instance variables.

D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Difference Between C++ and Java

Java	C++
Java doesn't support Pointer concept	It support pointer concept.
It doesn't support multiple inheritances.	It support multiple inheritance
Java does not include structures or unions.	It have structure and union concept.
Java includes automatic garbage collection.	C++ requires explicit memory management
Java has method overloading, but no operator overloading.	C++ supports both method overloading and operator overloading.
It is platform independent programming language	It is platform dependent programming language.
It is mainly used for design web based application but also use for develop desktop application.	It is used for design only desktop application like OS, Compiler etc.
Java uses compiler and interpreter both	C++ use only Compiler.
Java is high level programming language in java we write code like simple English language.	C++ is more nearer to hardware then Java

D Y PATIL DEEMED TO BE UNIVERSITY RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI


D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Thank You